xMCF
ISO/PAS 8329

Description of mechanical connections and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues #95 / #104
2026-11-04



Meeting results are added in green font. Additions are inserted in light blue.

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energizing great minds

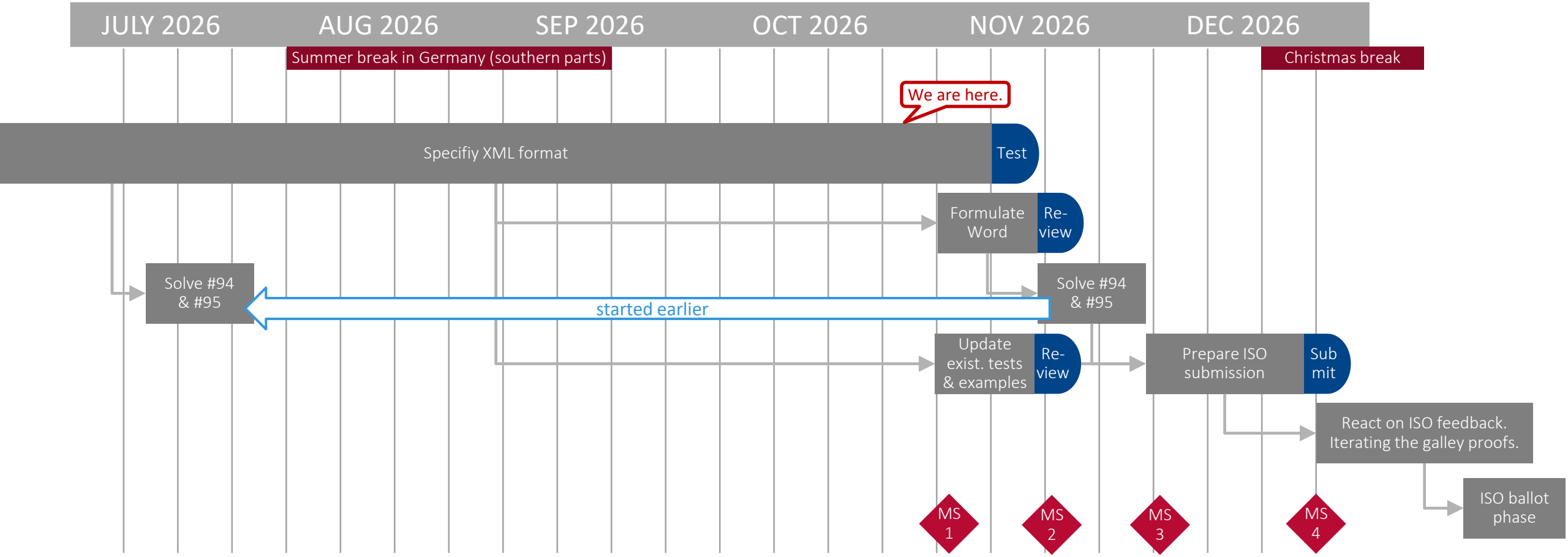
Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-11-04_VideoConference/README.md

- Welcome & Flying over today's topics
- Timeline for next release
- New attendee
- GH‘s comment to issue #105 of 2026-10-12
- Action Items Connected with Schematron
- Work in progress — focus on issues #95 / #94 et al.
- Cooperation Model (just for reference)
- Next meetings

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Timeline for the next release

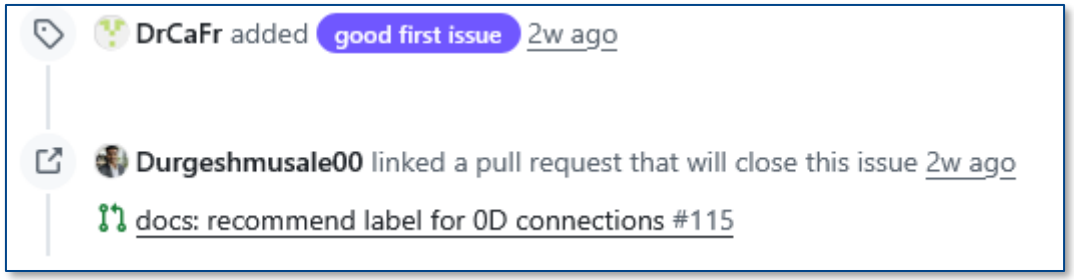


Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.

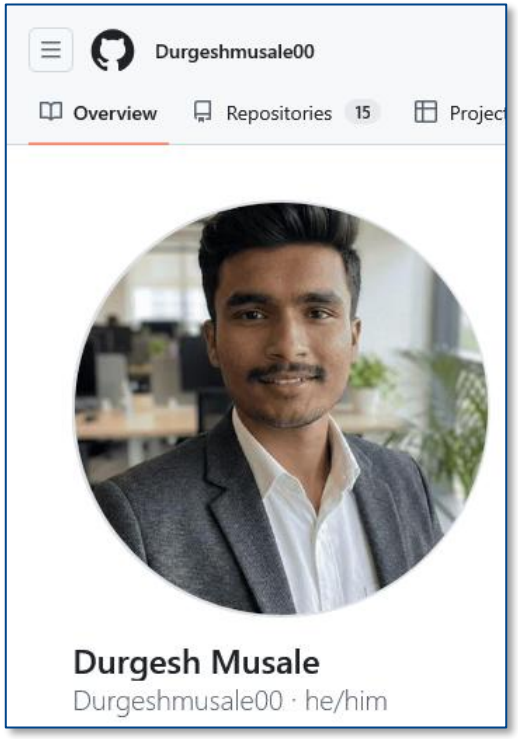
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Welcome to “Durgeshmusale00”

Looking at issue #113 , one encounters:



There is a profile at GitHub.



On 24 October, Durgesh indicated via LinkedIn that he would like to attend future meetings.

On 4 November, he introduced himself to the team.

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GH‘s comment to issue #105 of 2026-10-12

as of 2026-11-04

I tried to use the new XSD with the PDM-IF test datasets

1. although mentioned in a previous comment from NE, <connection_0d/> is commented out within <connection_templates/> and <connection_0/1/2d/> has no attribute "template_label".
   Use case: having 20 connections using the same rivet template and the same <custom_attributes/>, <femdata/>, ”quality_control“ (confirmed), ~~"connection_0/1/2d_type",~~ "contact_list“
   For "template_label", see slides “Screen Shot #1 & #2”, below. For custom attributes, see below.
   For femdata, templates do not make sense because contained information is always instance/location specific.
2. although mentioned in a previous comment from NE, I don't find <bolt_template/>, <nut_template/>, <screw_template/>, <washer_template/>, and have no attribute "template_label“,
   For "template_label", see slides “Screen Shot #1 & #2”, below.
3. Could we add "normal_direction" and "tangential_direction" to the templates?
   Use case : having 20 times the same rivet in the same direction.
   [*CF: → See our decision of 2026-04-29 and the rationale, why to not do so. / following slide*]
4. Could we allow "template into template", adding the attribute "template label" to the template objects?
   Use case: having 50 times the same rivet, 20 times in one direction and 30 times in another direction.
   [*CF: → See our decision of 2025-11-26 & 2025-12-10 and the rationale, why to not do so. / following slides*]

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Directions (normal & tangential) in templates?

as of 2026-04-29

Decision:

- *No* directions shall appear inside a connection template.

Rationale:

- The combination of several connection templates with a number *N* of different directions delivers a high number (Cartesian product) of combined templates to be managed and hence cuts down the desired repetition count by the corresponding factor *N* (on average).
  Consequently, this approach is neither scalable nor cost-effective.
- Idea (investigation postponed): Introduce *separate* templates for certain directions, e.g. "`negative_z`", *independent* from connection templates.

⇒ We keep this on the agenda for the later following χMCF release. ↦ issue #119

———————————————

- ~~Also to be investigated later:~~ Introduction of templates for `<custom_attributes/>`. Suggestion confirmed.
- Rationale: One cannot foresee, whether a custom attribute is instance-related or not. Thus, we can imagine valid use cases.
  But this validity cannot be defined by means of the standard. A customer specific Schematron file could be a means to address this.
- SH volunteers to change the XSD, accordingly.

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Proposed wording for results of our previous meetings
(2025-11-26 & 2025-12-10)

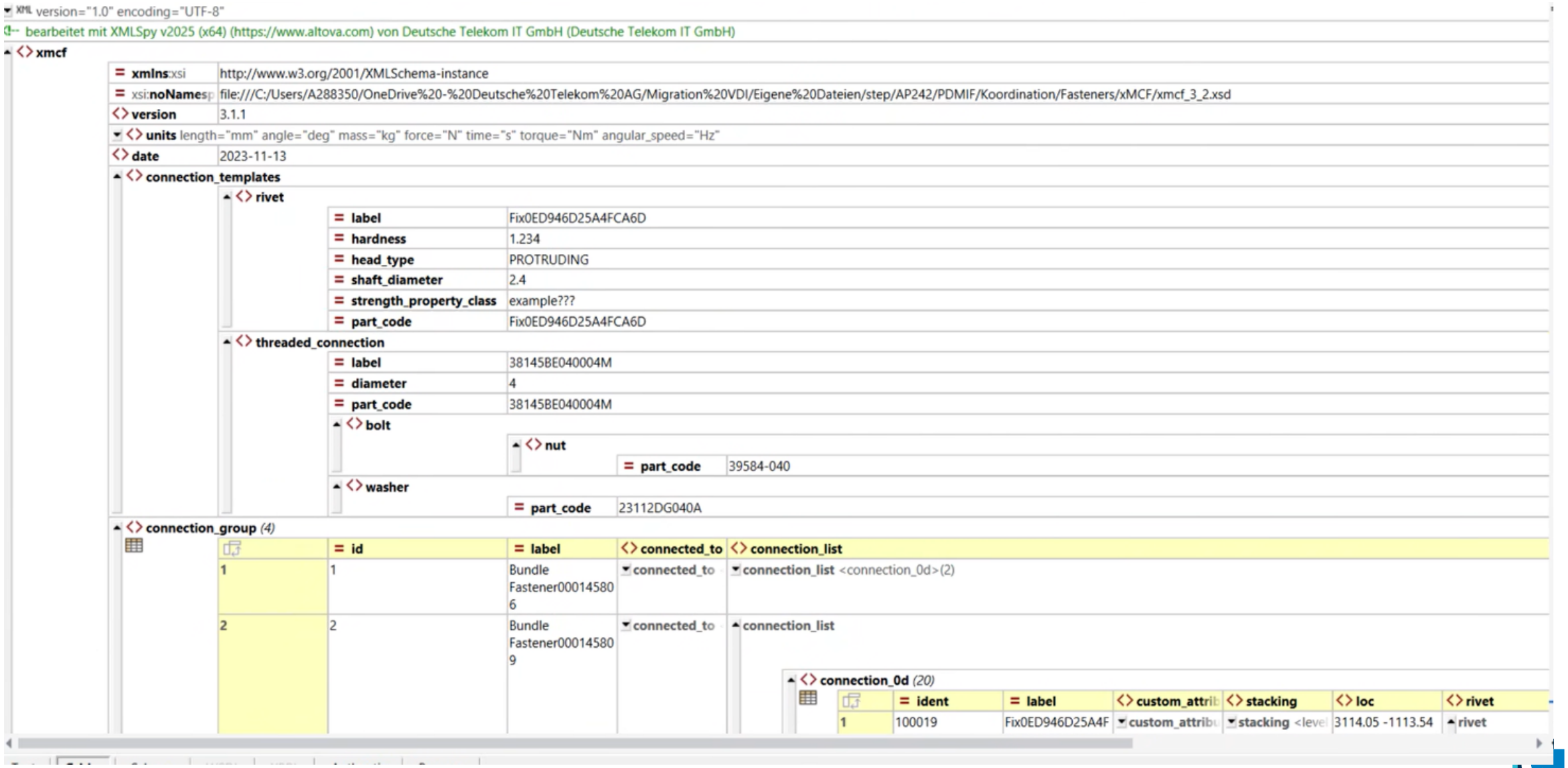
as of 2025-12-10

- If a connector in a χMCF file contains both,
  - a reference to catalog data,
  - and (some of) such data internally in its definition,
  the internal data shall take precedence.
- A connector may omit otherwise mandatory attributes if these can be determined via a reference to a catalog entry. XML schema validation must take this into account.
- Note (recursive references): A catalog entry *in χMCF format* (either internal or external file-based) may *not* (for the time being) refer to other catalog entries itself. The format of *database-based* catalog entries is not subject to the standard. Therefore, no statement is made here.
  At some point in the future, however, this feature could still prove useful for enabling the step-by-step completion of connector definitions.

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GH‘s comment to issue #105 of 2026-10-12 — Screen Shot #1

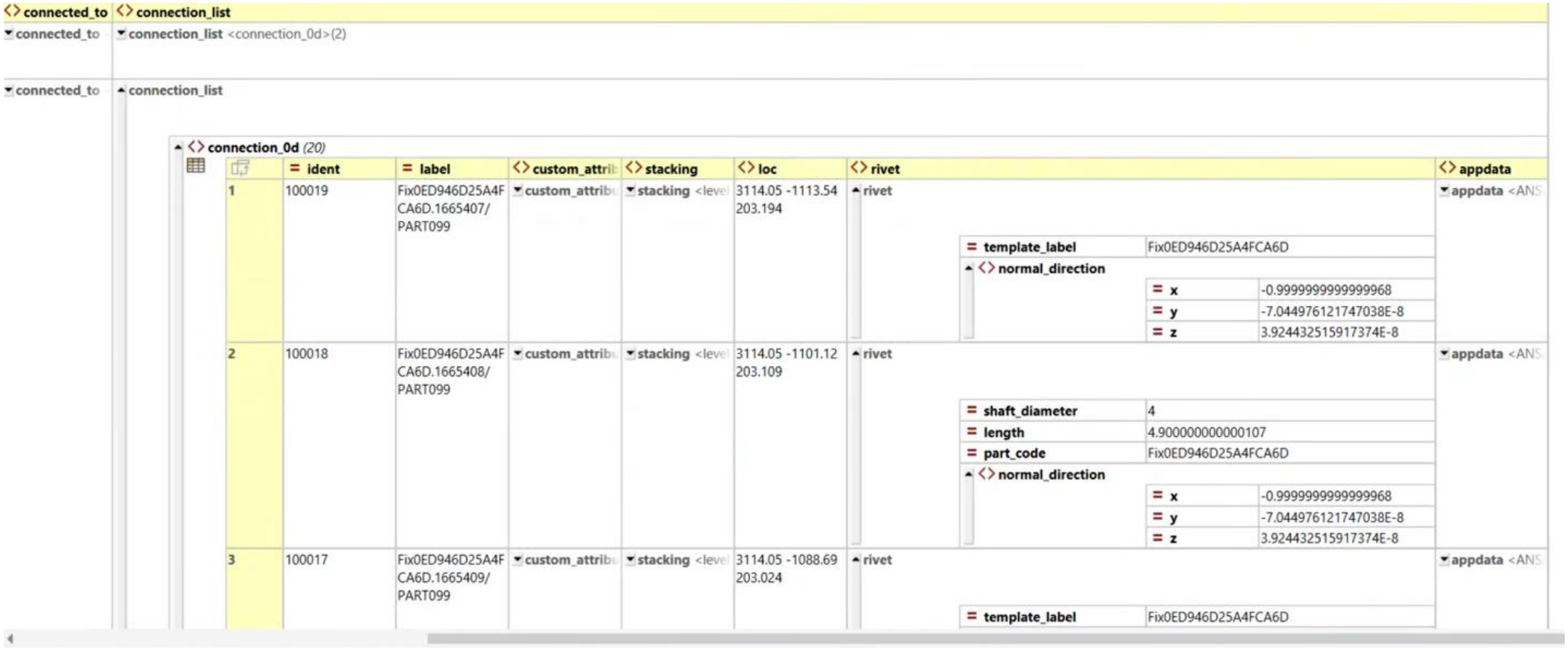
as of 2026-11-04



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GH‘s comment to issue #105 of 2026-10-12 — Screen Shot #2

as of 2026-11-04



connected_to | connection_list

connected_to | connection_list <connection_0d>(2)

connected_to | connection_list

connection_0d *(20)*

	ident	label	custom_attrib	stacking	loc	rivet	appdata
1	100019	Fix0ED946D25A4FCA6D.1665407/PART099	custom_attribu	stacking <level	3114.05 -1113.54 203.194	rivet	appdata <ANS
2	100018	Fix0ED946D25A4FCA6D.1665408/PART099	custom_attribu	stacking <level	3114.05 -1101.12 203.109	rivet	appdata <ANS
3	100017	Fix0ED946D25A4FCA6D.1665409/PART099	custom_attribu	stacking <level	3114.05 -1088.69 203.024	rivet	appdata <ANS

rivet (1)

template_label	Fix0ED946D25A4FCA6D
normal_direction	
x	-0.9999999999999968
y	-7.044976121747038E-8
z	3.924432515917374E-8

rivet (2)

shaft_diameter	4
length	4.900000000000107
part_code	Fix0ED946D25A4FCA6D
normal_direction	
x	-0.9999999999999968
y	-7.044976121747038E-8
z	3.924432515917374E-8

rivet (3)

template_label	Fix0ED946D25A4FCA6D

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Action Items Connected with Schematron

as of 2026-10-07

- CF to approach KLT for contributing some general intro about Schematron and a reference to STEP AP242.
  (Done. Result see next slide.)
- CF to extend distribution script by Schematron file, such that it appears next to the XSD on the ISO download server.
- We need a new general clause which describes methods & means for automated validation, like Java-based XSD- and Schematron validators.

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General intro about Schematron and a reference to STEP AP242

Provided by KLT via Mail, 2026-10-13. Quoted, here.

Schematron general presentation: (extracted from the presented slides)

The objective of using Schematron is to automatically verify XML documents against rules not covered by XSD language. Schematron is an international standard: ISO/IEC 19757-3:2020. Schematron is an XML rule-based validation language: It defines XPATH assertions on the structure and content of an XML document, that cannot be verified by an XSD check, and if the assertion fails, diagnostic messages are reported.

Schematron rules do not replace XSD validation but extend it with more constraint. A tool is required to execute Schematron rules files on XML files (by running XSL transformations provided by the standard) and to display the messages in text or HTML or other types of reports (using self-made XSL or else).

Connection to AP242:

STEP AP242 (ISO 10303-242) specifies information models and implementations schemas. Here we focus on the AP242 “Domain model” and to its XML implementation.

First, the AP242 Domain model is specified in SysML (object classes, attributes/properties, relationships and population constraint rules).
[*CF: SysML 1.4 is ISO/IEC 19514:2017*]

Then it is transformed to XSD and Schematron for implementation purpose: Object classes, attributes/properties and relationships are transformed to XSD, whereas population constraint rules are transformed to Schematron. These transformations are defined in ISO 10303-15 (another STEP standard).

Please let me know if it is ok for you.

Issue #95: Add persistent identifier to `connection_group`

as of 2026-07-01
≙ 2026-07-08

Not discussed
2026-10-02.

- Original request: „Up to now, `connection_groups` have an integer identifier which is local to each χMCF file, and not persistent.
  Under PDM considerations, an optional persistent alphanumerical id would help to increase the process reliability …"
- This use case has been very well justified.
- However, there are other use cases of great practical relevance where the introduction of a new mandatory attribute requiring attention would result in a significant additional workload, without this being offset by any benefit.
- The group decided to **continue the discussion** at the **next meeting** with the question:
  'What if the attribute (persistent identifier) were just *optional*?' → See next slide.

- The following important point was also raised during the discussion:
  `<connected_to/>` lists the connected parts in *no particular order*.
  The order may change even after a simple im- / export sequence.
  If the stacking sequence is important, always use the `<stacking/>` element!
  → 2026-09-06: An according issue #112 was created.

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Issue #95: Add persistent identifier to `connection_group`

(continued) 'What if the attribute (persistent identifier) were just *optional*?'

Results of 2026-07-22 & 2026-07-28

Not discussed 2026-10-02.

- PDM Use Cases: Exchange with Partner and recognize a known connection group upon re-import.
- CAD Use Cases: Creating, Deleting, Splitting, Merging, Editing of connections, incl. changing the connected parts.
- PEP is "owner" of the unique connection group IDs, just as with the part codes.
  (Single Point of Truth to guarantee uniqueness!)
  PDM system, as "incarnation" of the PEP, is the tool which creates and keeps authority on such IDs.
- Difference between part codes and connection group IDs:
  For CAD data, there is (usually) only *one* CAD authoring system, which is tightly integrated with PDM.
  For connection data, this frequently is different:
  Many tools are established for manipulating connections, which are not tightly PDM-integrated.
- Solution ideas:
  a. PDM system offers and tool attaches to a standardized interface for querying suitable connection group ID.
  b. Tools could exchange one χMCF file per connection group
  (which renders the connection group ID as meaningless, as it is currently.)
  c. More ideas? --- no → Concept is confirmed (2026-09-16). ➔ Text for spec update to be drafted in comments to issue #95.
- Integer connection group *id* shall be supplemented with an alphanumeric *label*.
  2026-09-09: ⇒ XSD to be updated. / 2026-09-16: SH added according change to GitHub, this morning.

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Issue #94: Add instance identifier to a connector

as of 2026-09-09

Not discussed 2026-10-02.

- Original request: „In order to keep consistency between the PDM system and the connection data, it is required to have an instance id for the rivets“ (Rivets or any other connection method in which a separate physical component establishes the connection.)
- The standard documents says in clause “8.2.2 Attribute `label`”: “Any connection should have an attribute called `label`, which identifies it throughout the entire CAx process, maybe even throughout the complete product lifecycle, including manufacturing. It is not necessary that these labels are unique.”
  Remark 1: The use cases in which “`label`” is not unique are quite specific and rare.
  Remark 2: “`label`” must not be confused with “`part_code`” , which frequently is *not* unique. Example:

```
<connection_0d label="BOLT_135" part_code="BOLT-NUT-Combi #17"> …  <!-- only label is unique -->
  <threaded_connection … part_code="M10x50 8.8"/>   <!-- not unique -->
</connection_0d>
<connection_0d label="BOLT_136" part_code="BOLT-NUT-Combi #17"> …  <!-- only label is unique -->
  <threaded_connection … part_code="M10x50 8.8"/>   <!-- not unique -->
</connection_0d>
```

In PDM, e.g. `label="BOLT_135"` would denote an *instance* combination of e.g. a bolt, some washers and a nut.
However, a part code *for this combination* is needed and not yet supported by χMCF.
⇒ Suitable place for the part_code for the combination to be found in χMCF syntax – at both, the actual connection plus its template, if there is one.
Suggestion: Put the part_code as shown above in blue, at both positions.
— ***No***, it should be an attribute of `<threaded_connection/>`, see next slide!
2026-09-09: For discussion & validation, see following slides!

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#94 wrt. templates

as of 2026-09-09

Not discussed 2026-10-02.

- The combination's part_code should *not* reside on `<connection_0d/>`, because there it would also be accessible for e.g. spotwelds.
  Instead, it should be an attribute of `<threaded_connection/>`.
  Thus, another name is needed for the combination's part_code, namely "`combination_part_code`".
- In case a connection template is used, the part_code of a connection may be given there.
- The minimal content of a <connection_0d/> would then be:
  - Label (highly recommended)
  - Template (as mentioned above)
  - <loc/>
- The part code of the connection (e.g. the bolt) and its components (screw, nut, washers) are then given only once,
  i. e. in the template definition.
- Remark: In any case, if there are different components of connections with the same combination's part code,
  this is an inconsistency and must be prevented by the exporting system.
  ⇒ Schematron rule to be added.
- PDM Systems would differentiate e.g. Screw-Nut-Combinations according to the distance between screw head and nut.
  Since χMCF does not cover this kind of distance, as more general geometric parameters too, tools cannot detect this situation.
- To-Dos: Extend XSD. Extend the Schema. Provide example file(s): With & without template. Formulate a text fragment for the spec.

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Add this to the χMCF Specification

Not discussed 2026-10-02.

Label is optional,
but highly recommended,
especially in cases,
when different tools or
partners are involved.

2026-09-15: An according issue #113 was created.

9.1.1.1 General

The possible XML attributes of the <connection_0d/> element, which describes a point connection (0D connection), are specified in Table 32:

Table 32 — Attributes of element <connection_0d/>

Attributes	Type	Use	Constraint
label	Alphanumeric	Optional	-
ident	Integer	Optional	positive, unique within a χMCF file
quality_control	Alphanumeric	Optional	See clause 8.4.

9.1.1.2 Attribute "label"

The label defines the human readable identification of a connection. It can contain a description of the connection or simply an index as an integer.

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Other (ISO) Standards

resumed from 2025-11-26 / 2025-12-10 / 2026-01-14

Not discussed 2026-10-02.

#61 suggests to analyze existing standards in this context, such as

- ISO 13584-511:2006 — Industrial automation systems and integration — Parts library — Part 511: Mechanical systems and components for general use — Reference dictionary for fasteners
  — a copy was provided to the WG — for internal use, only.
- ISO 1891:2009 — Fasteners — Terminology
  — a copy was provided to the WG — for internal use, only.

  A sincere *thank you* to all donors!

- 2026-09-15: An according issue #114 was created for this analysis.

- Do we need / want to consider more standards?

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Cooperation Model (just for reference)

as of 2026-10-07

Next steps in expanding the standard text and how we coordinate our joint work on this

- For issues #61 and #105, there is a shared solution via “Templates”, so they should be worked on together.
- Work on issue #95 can proceed independently of this. Same with #94.
- In general, we need for each of these
  - Some words for the change log
  - New / changed clause(s) with explanation & motivation of the use case(s) and the solution concept.
  - Description and example files for the syntax
  - test cases / test files for the syntax
- “Small” issues #112 … #118 can be picked by anyone, any time.

Modus operandi:

- The one who starts working on a binary file (especially the Word file) places a *semaphore* file "${filename}_locked_by_XYZ.txt" next to the binary in GitHub, such that we can synchronize changes and avoid later merge conflicts.
- Let’s focus on content, first, not on formatting. Especially page breaking is postponed.
- We can have separate branches for #94 and #95 (if necessary).

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Next meetings — Suggested new appointments

- 2026-11-11, 15:00 CET — focus group on #105 / #61 — NE, GH & HS declined. Only SH confirmed.
  ➔ review of SH's suggestions.
  — objective: agree on XML suggestions & examples / draft Word file changes

- 2026-11-18, 15:00 CET — focus group on #105 / #61
  — objective: agree on XML suggestions & examples / draft Word file changes
- 2026-12-02, 15:00 CET — focus group on #105 / #61
- 2026-12-16, 15:00 CET — focus group on #105 / #61
- 2027-01-13, 15:00 CET — focus group on #105 / #61

➔ confirmed

CF sent / will send out invitations.

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